

Model We are asked to find the magnetic field due to a simple current distribution, so this example is a typical problem for which the Biot–Savart law is appropriate. We must find the field contribution from a small element of current and then integrate over the current distribution from θ_1 to θ_2 , as shown in **Figure 29.3b**.

Analyse Let's start by considering a length element $d\vec{s}$ located a distance r from P . The direction of the magnetic field at point P due to the current in this element is out of the page because $d\vec{s} \times \hat{r}$ is out of the page. In fact, because *all* the current elements $I d\vec{s}$ lie in the plane of the page, they all produce a magnetic field directed out of the page at point P . Therefore, the direction of the magnetic field at point P is out of the page and we need only find the magnitude of the field. We place the origin at O and let point P be along the positive y axis, with $\hat{k}$ being a unit vector pointing out of the page.

From the geometry in **Figure 29.3a**, we can see that the angle between the vectors $d\vec{s}$ and $\hat{r}$ is $\left(\frac{\pi}{2} - \theta\right)$ radians.

Evaluate the cross product in the Biot–Savart law:

$$d\vec{s} \times \hat{r} = |d\vec{s} \times \hat{r}| \hat{k} = \left[dx \sin\left(\frac{\pi}{2} - \theta\right) \right] \hat{k} = (dx \cos \theta) \hat{k}$$

Substitute into **Equation 29.1**:

$$d\vec{B} = (dB) \hat{k} = \frac{\mu_0 I}{4\pi} \frac{dx \cos \theta}{r^2} \hat{k} \quad (1)$$

From the geometry in **Figure 29.3a**, express r in terms of θ :

$$r = \frac{a}{\cos \theta} \quad (2)$$

Notice that $\tan \theta = -x/a$ from the right triangle in **Figure 29.3a** (the negative sign is necessary because $d\vec{s}$ is located at a negative value of x) and solve for x :

$$x = -a \tan \theta$$

Find the differential dx :

$$dx = -a \sec^2 \theta d\theta = -\frac{a d\theta}{\cos^2 \theta} \quad (3)$$

Substitute **Equations (2) and (3)** into the magnitude of the field from **Equation (1)**:

$$dB = -\frac{\mu_0 I}{4\pi} \left(\frac{a d\theta}{\cos^2 \theta} \right) \left(\frac{\cos^2 \theta}{a^2} \right) \cos \theta = -\frac{\mu_0 I}{4\pi a} \cos \theta d\theta \quad (4)$$

Integrate **Equation (4)** over all length elements on the wire, where the subtending angles range from θ_1 to θ_2 as defined in **Figure 29.3b**:

$$B = -\frac{\mu_0 I}{4\pi a} \int_{\theta_1}^{\theta_2} \cos \theta d\theta = \frac{\mu_0 I}{4\pi a} (\sin \theta_1 - \sin \theta_2) \quad (29.4)$$

Check the dimensions, noting that the quantity in brackets is dimensionless:

$$[\text{MQ}^{-1}\text{T}^{-1}] = [\text{MLQ}^{-2}][\text{QT}^{-1}]/[\text{L}] = [\text{MQ}^{-1}\text{T}^{-1}] \quad \text{☺}$$

(B) Find an expression for the field at a point near a very long current-carrying wire.

Solution

We can use **Equation 29.4** to find the magnetic field of *any* straight current-carrying wire if we know the geometry and hence the angles θ_1 and θ_2 . If the wire in **Figure 29.3b** becomes infinitely long, we see that $\theta_1 = \pi/2$ and $\theta_2 = -\pi/2$ for

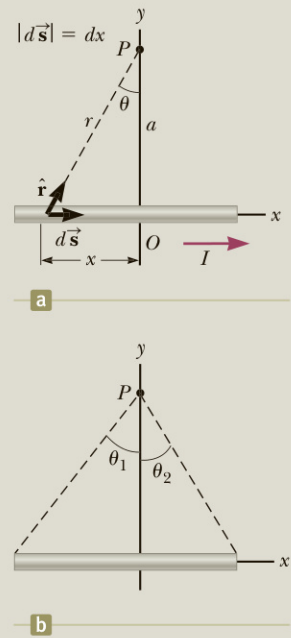


Figure 29.3
(Example 29.1) (a) A thin, straight wire carrying a current I (b) The angles θ_1 and θ_2 are used for determining the net field.